

- precision: determined by the range of values/measurements B1
- 2 (a) Straight line graph through origin, so A1
 extension is proportional to applied force, it obeys Hooke's law

(b) area under graph = $\frac{1}{2} Fx = \frac{1}{2} (12.0)(8.0 \times 10^{-2})$ A1
 $= 0.48 \text{ J}$

Comments: Students must write down the appropriate algebraic equation and then show the full substitution of numerical values.

(c) (i) conservation of momentum, $4.0 v_A = 6.0 v_B$ M1

$$\text{ratio} = \frac{0.50 \times 4.0 \times v_A^2}{0.50 \times 6.0 \times v_B^2} = \frac{0.50 \times 4.0}{0.50 \times 6.0} \left(\frac{6.0}{4.0} \right)^2$$

C1

$$= 1.5$$

A1

Comments: The force was not constant and the acceleration was not uniform. It was wrong to use kinematics equations ($v = u + at$ or etc) to work out the answer. No credit for such an approach.

(ii) $0.48 = (\text{KE of A}) + (\text{KE of B})$

$$0.48 = \frac{5}{3} (\text{KE of A}) = \frac{5}{3} \left(\frac{1}{2} \times 4.0 \times v_A^2 \right) \quad \text{C1}$$

$$v_A = 0.38 \text{ m s}^{-1} \quad \text{A1}$$

(d) curve starts from origin and has decreasing gradient B1

final gradient of graph is zero B1

Comments: The graph is the first quarter of a sine graph. Few students understood and showed clearly that when the extension of the spring becomes zero the force on block A will become zero so that the gradient of the graph (rate of change of velocity) becomes zero momentarily. The graph should stop once gradient has reached zero.

(e) Momentum of A and momentum of B are always equal in magnitude
but opposite in directions (conservation of momentum) M1

So the two masses must oscillate with the same frequency. A1

3 (e) $T \sin \theta = ma$

$\uparrow$ T , tension in string